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Lattice clustering and its application in credit risk management of commercial banks

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Abstract

One of the classical methods of clustering is kNN and it is a simple and widely used algorithm. But when it comes to the samples with unstructured and fuzzy values, it is not applicable enough course it is based on the Euclidean distance, which does not have practical significant for such data . But these kinds of data are much important in credit risk management of commercial banks, therefore we must do a deal with these important variables by fuzzy division, and calculate the fuzzy distance between every two samples within the lattice degree, then generate a fuzzy division matrix to replace the Euclidean Distance in kNN algorithm.

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1. Introduction

Approaches to clustering have been widely used in many fields of scientific research and practical application such as Internet, big data and information system. Most algorithms of clustering, such as kNN algorithm, are based on the criteria of distance, especially Euclidean distance. However, when it comes to clustering the samples that are represented by fuzzy variables, Euclidean distance is not suitable enough, for it doesn't have practical significance for fuzzy variables.

This paper aims to solve this problem; fortunately, we found that using fuzzy distance in clustering is such a great solution. On the basis of this, we propose a new method of clustering: **Lattice Clustering**. We use **lattice nearness degree** to denote the fuzzy distance, which is related to scalar product and cross product and describes the closeness

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between samples. Then we replace Euclidean distance with lattice nearness degree in our clustering practice, therefore the distance has the theoretical and practical significance.

The rest of the paper is structured as follows: The next part is the literature review that is about clustering, kNN and lattice. Section 3 is the main method of this paper, which concludes the definition of lattice and calculation formulas. Then we display the result of kNN and compare it with the result of lattice clustering. The last part is the conclusion and our future work.

2. Literature Review

Among popular approaches to clustering, the classic K Nearest Neighbor (kNN) algorithm remains a standard tool with good performance characteristics and short training time, also it was demonstrated at a high level in earlier researches [1,2]. Because of its superior performance and simple implementation, the one of the oldest methods for pattern classification has been widely used in many fields such as data mining, machine learning and etc. [3,4] In addition, the metric for calculating the distance between different examples heavily affects the accuracy of it [5,6]. We can also apply kNN to infer the tags of pictures from pictures shared by users and associated tags [7].

However, kNN algorithm has its disadvantages as well. For example, no weight is assigned to neighbors and It is vulnerable to data imbalances [2]. In spite of many optimal methods of kNN algorithms has been researched out [8,9], it can not discuss samples denoted by fuzzy variables now.

Then we introduced the concept of 'lattice' to the clustering to solve the problem. Lattice rules are quadrature rules for the numerical approximation of the s-dimensional integral [10] and an instance of it is number-theoretic rules [11]. Lattice also can be used in many fields [12] such as crypto-systems [13]. But in this paper, we try to find out whether it is a more accurate method of clustering to replace the distance in kNN by lattice degree.

3. Main methods

3.1. The definition of lattice

We have to know the concept of the special partially ordered set at first, cause lattice is a special set of that. Let R be a relation on the set A . If R is reflexive, anti-symmetric and transitive, then R is called a partially ordered relationship of A , and recorded as ' $\leq$ '. For $(a,b) \in R$, it is expressed as $a \leq b$. Giving a partially ordered relationship ' $\leq$ ' on the set A , then the set A is called to form a partially ordered set according to the partially ordered relationship ' $\leq$ ', and the set A with the partially ordered R is called a partially ordered set, which is denoted as $(A, \leq)$.

Giving a partially ordered set $(A, \leq)$. If any two elements in A have their supremum and infimum, then we regard $(A, \leq)$ is lattice.

3.2. The formula of lattice nearness degree

Let the domain $X = \{X_1, X_2, \dots, X_n\}$, where X_i ($i=1,2,\dots,n$) expresses the variable of X . A and B are two fuzzy sets of X , and $A = \{(X_1, x_{1a}), (X_2, x_{2a}), \dots, (X_n, x_{na})\}$, $B = \{(X_1, x_{1b}), (X_2, x_{2b}), \dots, (X_n, x_{nb})\}$, where x_{ia} and x_{ib} ($i=1,2,\dots,n$) separately represents the membership degrees of A and B of X_i , and the values of x_i are between 0 and 1.

Then we denote lattice nearness degree as $\sigma_L(A, B)$ between A and B , and the formula is as follow:

$$\sigma_L(A, B) = \frac{A \bullet B + (1 - A \otimes B)}{2},$$

where $A \bullet B$ and $A \otimes B$ separately represent the scalar product and the cross product, and the formulas are as follows:

$$A \bullet B = \bigvee_{x \in X} (A(x) \wedge B(x)), A \otimes B = \bigwedge_{x \in X} (A(x) \vee B(x)).$$

3.3. Other related work

In addition to the definition and the formula, we must do some other previous work before lattice clustering, such as screening of samples and variables. We selected them from an excel of credit data about borrowers' information from a financial company, and finally we chose 4 credit variables that impacted most significant according to the results of correlation analysis: 'bank_account', 'bank_legal_num', 'bank_org_num' and 'credit_legal_org_num', respectively representing the total number of 'credit count', 'credit institutions', 'credit corporation' and 'card issuer'. We use X_1, X_2, X_3, X_4 to express the variables and $x_{1,i}, x_{2,i}, x_{3,i}, x_{4,i}$ ($i=1,2,\dots,200$) to express the values of each variable. Then we eliminate the outliers and finally filtered 250 records of borrowers from the rest of data and we use $S_1, S_2, \dots, S_{200}$ to represent them.

The following work was dealing with the values of variables, and the target of this step is to let each value change to the degree of membership that is between 0 and 1. Here's how to do it: Divide each value of variable by the maximum value of this variable, and the results are the value we want. So the expression of fuzzy set S_i ($i=1,2,\dots,200$) can be expressed like this: $S_i = \{(X_1, x'_{1,i}), (X_2, x'_{2,i}), (X_3, x'_{3,i}), (X_4, x'_{4,i})\}$. More details are in the 'Table 1'.

Table 1. Details of the Variables

Variables	Variables Attributes	Range of Values		Procession Method
		Max	Min	
X_1 : bank_account	Discrete value in positive integers	12	1	$x'_{1,i} = x_{1,i}/12$
X_2 : bank_legal_num		8	1	$x'_{2,i} = x_{2,i}/8$
X_3 : bank_org_num		9	1	$x'_{3,i} = x_{3,i}/9$
X_4 : credit_legal_org_num		7	0	$x'_{4,i} = x_{4,i}/7$

Lattice clustering uses the degree to cluster the data-set, where there is this method of clustering creative and different from kNN.

3.4. The steps of lattice clustering

Step 1: Determine two cluster centers

This is the start and the key point of lattice clustering, and it lays the foundation for the later steps cause it has a direct impact on the clustering results. We selected 2 virtual points as centers of 2 clusters according to the frequency of each variable, one of which was made up of the modes (the value that appears most frequently in a variable) and the values of the other were exactly the opposite values of the modes. The former center means the traditional borrower that is most typical and we denote it as $C_{\max}$, and the latter is the most active borrower that is denoted as $C_{\min}$. They are represented as follows:

$$C_{\max} = \{(X_1, x'_{1,\max}), (X_2, x'_{2,\max}), (X_3, x'_{3,\max}), (X_4, x'_{4,\max})\},$$

$$C_{\min} = \{(X_1, x'_{1,\min}), (X_2, x'_{2,\min}), (X_3, x'_{3,\min}), (X_4, x'_{4,\min})\}.$$

However, there existed 1 in $x_{i,\min}$ ($i=1,2,3,4$), to ensure the dimensional consistency, we replaced 1 with the value of the second last frequency in the same variable, and cut the corresponding data, then we filtered 215 records and denoted them as S'_i ($i=1,2,\dots,187$).

Step 2: Calculate the $\sigma_L(S'_i, C_{\max})$ and $\sigma_L(S'_i, C_{\min})$

We use the formula of $\sigma_L(A, B)$ as mentioned above to calculate the lattice nearness degrees between each sample S'_i ($i=1,2,\dots,187$) with $C_{\max}$ and $C_{\min}$ separately denoted as $\sigma_L(S'_i, C_{\max})$ and $\sigma_L(S'_i, C_{\min})$.

Step 3: Cluster the data-set

According to the results in step 2, we can find the minimum between the two values of each sample and represent it as $\sigma_{L,i}$ ($i=1,2,\dots,187$), and on the basis of $\sigma_{L,i}$, each sample can be classified to the cluster the center of that is closer to the sample, then 187 samples are divided into 2 clusters, where cluster 1 (the samples in this cluster are closer to $C_{\max}$) includes 53 data points, and the other (the samples in this cluster are closer to $C_{\min}$) includes 134 points. The distribution of lattice degree is as 'Figure 1' below, in which 'L1' and 'L2' respectively represent $\sigma_L(S'_i, C_{\min})$ and $\sigma_L(S'_i, C_{\max})$. So we can find that the points above the dividing line are divided into cluster 1 and them below the line belong to cluster 2.

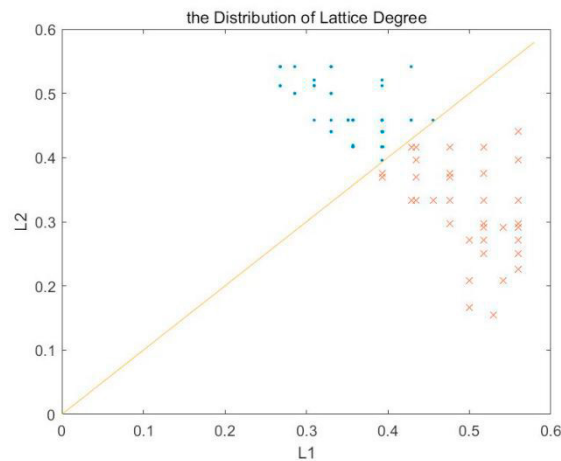


Fig. 1. The Distribution of Lattice Degree

Step 4: Correlation test and label

The last step of lattice clustering is a correlation test between the set of $\sigma_{L,i}$ ($i=1,2,\dots,187$) and X_i ($i=1,2,3,4$). This test aims to find two variables that mainly impact the set of $\sigma_{L,i}$, and the results shows that 'bank_account' and 'bank_legal_num' are two key variables that impact $\sigma_{L,i}$ most. The results are in the 'Table 2'.

Table 2. The Results of Correlation Analysis (Pearson Coefficient)

	X_1	X_2	X_3	X_4
$\sigma_{L,i}$	0.680	0.579	0.594	0.556

From the table, we can easily find that X_1 and X_3 are the most significant correlation to $\sigma_{L,i}$, so we can say that the two variables have the most influence on lattice degree.

4. Comparison

In this section, we will make a comparison between lattice clustering and kNN. We used the app ‘Classification Learner’ in MATLAB to cluster the 187 records mentioned in 3.4 with the original values by the means of different kinds of kNN, and the results are as follows:

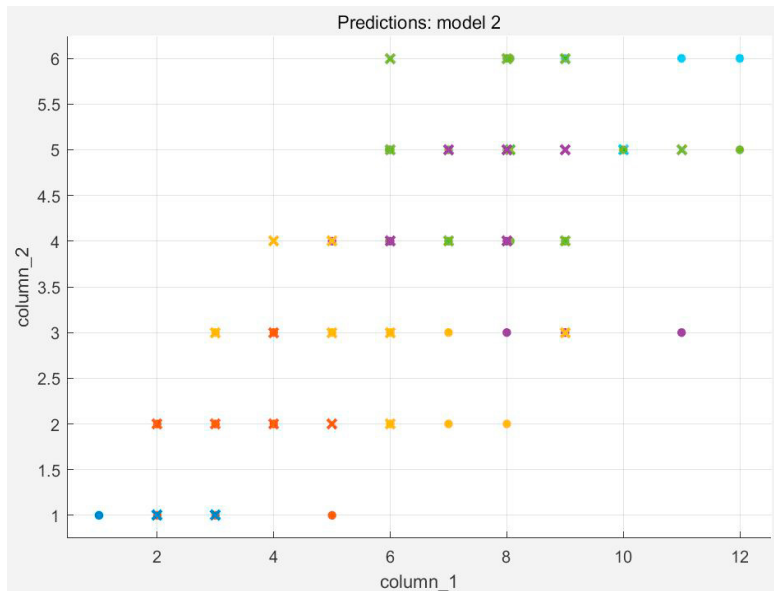


Fig. 2. Results of Fine KNN

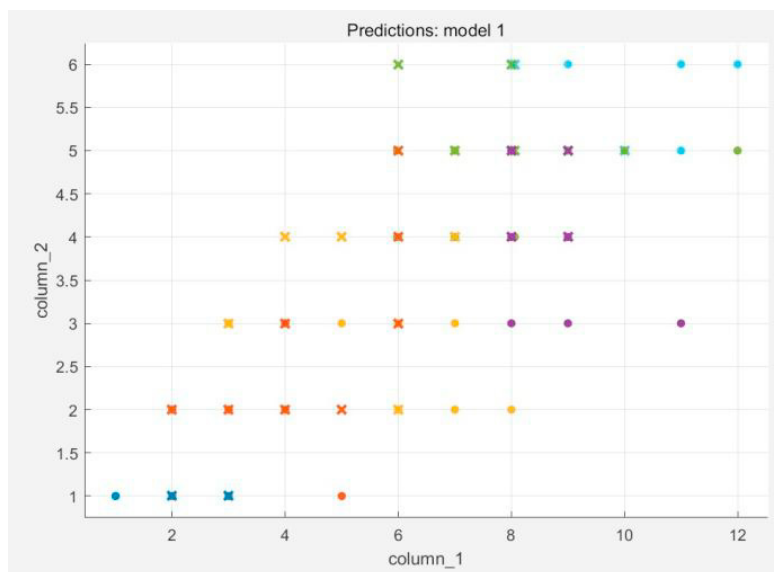


Fig. 3. Results of Medium KNN

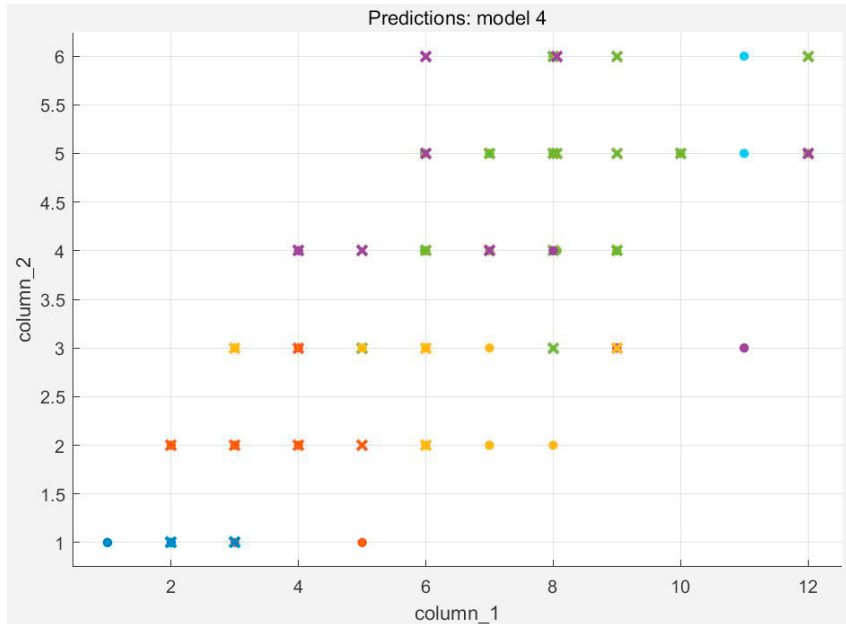


Fig. 4. Results of Coarse KNN

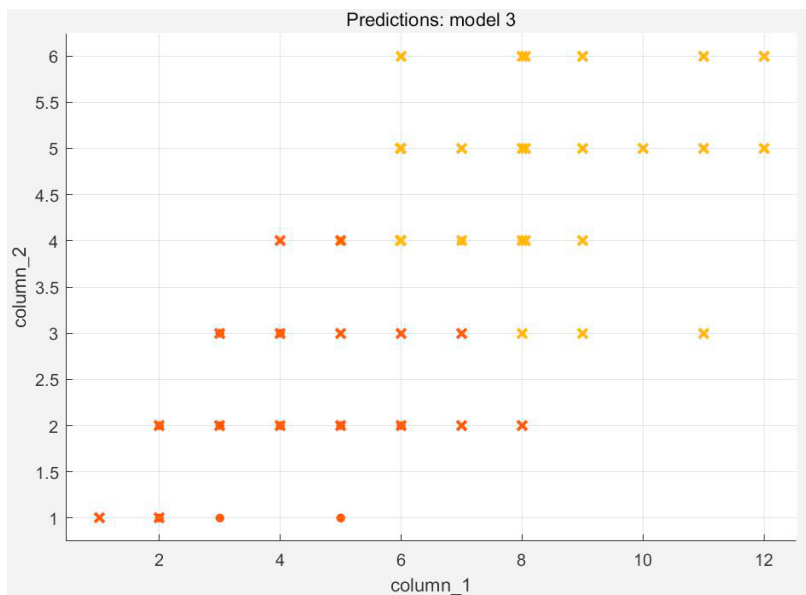


Fig. 5. Results of Cosine KNN

The ‘.’ in the figures represent the samples that are correctly clustered and the ‘×’ represent the incorrect ones, and more color in the figure means the results of clustering are worse. So from these figures, we can see that the results of different kNN are unsatisfactory, and the best result is Fine KNN with about 63% accuracy and the worse is Coarse KNN with 24% accuracy.

5. Conclusion and future work

In this paper, we created a new method of clustering with lattice degrees of samples. Compared to kNN, we found that for unstructured data, Lattice Clustering has better effect. It replaced the Euclidean distance in kNN with lattice degree and solved the problem that the distance is not applicable enough to unstructured and fuzzy data for it has no practical significance, in addition, the results obtained using this method of lattice clustering is more intuitive than the kNN. Lattice clustering filtered the outliers and easily divided the rest into two clusters, which is more effective and practical for credit risk management.

Our future work is further research on lattice to optimize the algorithm of lattice clustering. We will study it mathematically to make it more reasonable for the process of unstructured numbers. We also will do more tests on other classical clustering methods such as k-means and the emerging methods like DBSCAN with lattice degree, and finally write a complete algorithm with lattice degree specially for unstructured and fuzzy data.

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